

```

function forward_difference(f, x, h)
    return (f(x + h) - f(x)) / h
end

function backward_difference(f, x, h)
    return (f(x) - f(x - h)) / h
end

function centered_difference(f, x, h)
    return (f(x + h) - f(x - h)) / (2h)
end

```

centered_difference (generic function with 1 method)

```

function richardson_extrapolation_forward(f, x, h; es=1e-4, max
    approx_prev = NaN           # Will hold the previous ext
    errors = Float64[]          # To store the successive ap
    levels = Int[]              # To record the Richardson l
    approx_current = 0.0

    for level in 1:max_levels
        # Build the Richardson table for the current level
        D = zeros(level, level)

        # Column 1: forward difference at multiple step sizes
        for i in 1:level
            hi = h / 2^(i-1)
            D[i, 1] = (f(x + hi) - f(x)) / hi
        end

        # Richardson table fill (FORWARD => factor = 2^(j-1))
        for j in 2:level
            for i in j:level
                D[i, j] = (2^(j-1) * D[i, j-1] - D[i-1, j-1]) /
            end
        end

        approx_current = D[level, level]

        if level > 1
            error_est = abs(approx_current - approx_prev) * 100
            push!(errors, error_est)
            push!(levels, level)
            println("Level $level: Approximation = $approx_curr

            if error_est < es

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        println("Tolerance reached at level $level.")
        return approx_current, level, errors, levels
    end
else
    push!(errors, 100.0)
    push!(levels, level)
    println("Level $level: Approximation = $approx_curr")
end

approx_prev = approx_current
end

println("Maximum levels reached without meeting tolerance.")
return approx_current, max_levels, errors, levels
end

```

richardson_extrapolation_forward (generic function with 1 method)

```

# f1(x) = 0.25sin(x)
# f2(x) = 80 + 40cos(x)
# f3(x) = 30e^x
# x is time

```

```

f1(x) = 0.25 * sin(x)
f2(x) = 80 + 40 * cos(x)
f3(x) = 30 * exp(x)

```

f3 (generic function with 1 method)

```

x0 = 0.0
h = 0.5
es = 1e-4          # specific error < 0.0001
max_levels = 20    # เพื่อให้ถึง tolerance แน่ๆ

```

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```
approx1, level1, errors1, levels1 = richardson_extrapolation_forwar
println("\nFinal rate (f1') at x0 = $x0 : ", approx1)
```

```
Level 1: Approximation = 0.2397127693021015
Level 2: Approximation = 0.25509514920694437, Successive Error = 6.0
Level 3: Approximation = 0.25002824931286893, Successive Error = 2.0
Level 4: Approximation = 0.24999799470309764, Successive Error = 0.0
Level 5: Approximation = 0.249999985410959, Successive Error = 0.00
Level 6: Approximation = 0.25000000002343936, Successive Error = 5.9
Tolerance reached at level 6.
```

```
Final rate (f1') at x0 = 0.0 : 0.25000000002343936
```

```
approx2, level2, errors2, levels2 = richardson_extrapolation_forwar
println("\nFinal rate (f2') at x0 = $x0 : ", approx2)
```

```
Level 1: Approximation = -9.79339504877018
Level 2: Approximation = -0.15463000382354153, Successive Error = -6
Tolerance reached at level 2.
```

```
Final rate (f2') at x0 = 0.0 : -0.15463000382354153
```

```
approx3, level3, errors3, levels3 = richardson_extrapolation_forwar
println("\nFinal rate (f3') at x0 = $x0 : ", approx3)
```

```
Level 1: Approximation = 38.92327624200769
Level 2: Approximation = 29.242823763050268, Successive Error = 33.1
Level 3: Approximation = 30.023335371713426, Successive Error = 2.59
Level 4: Approximation = 29.999713930089992, Successive Error = 0.07
Level 5: Approximation = 30.000001462937284, Successive Error = 0.00
Level 6: Approximation = 29.99999996784982, Successive Error = 4.88
Tolerance reached at level 6.
```

```
Final rate (f3') at x0 = 0.0 : 29.999999996784982
```

```
println("\n==== SUMMARY (Richardson Forward, x0=0, h=0.5, es<0
println("f1'(0) = ", approx1, " (level ", level1, ")")
println("f2'(0) = ", approx2, " (level ", level2, ")")
println("f3'(0) = ", approx3, " (level ", level3, ")")
```

```
==== SUMMARY (Richardson Forward, x0=0, h=0.5, es<0.0001) ====
f1'(0) = 0.25000000002343936 (level 6)
f2'(0) = -0.15463000382354153 (level 2)
f3'(0) = 29.999999996784982 (level 6)
```